CS {4/6}290 & ECE {4/6}100 - Spring 2021

Homework 1: Performance Evaluation

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Due: January 29th, 2021 @ 11:59 PM

Rules

- **This is an individual assignment. ABSOLUTELY NO COLLABORATION IS PERMITTED**. All cases of honor code violations will be reported to the Dean of Students.
- The due date at the top of the assignment is final. Late assignments will not be accepted.
- Please use office hours for getting your questions answered. If you are unable to make it to office hours, please email the TAs.
- Make sure you **show your work and give explanations** for all questions unless otherwise stated.
- All answers must be **handwritten or typed**, and an electronic copy (PDF format) uploaded to Canvas before the due date.
- Make sure your answers are written within the boxes we have provided. We will be using Gradescope for grading and answers outside the boxes will be recognized by the system.

Question 0 – Short Answer Questions

1. According to Flynn's Taxonomy of classifying instructions and data that are processed by a machine, what would be the best way to classify the current generation of x86 Intel processors? **Please state your reasoning.**

We can classify current generation of x86 Intel processors as **MIMD.** Because today's processors mostly have more than one cores/PEs, and that each core can operate at different clock speed, work on different threads asynchronously. These are MIMD characteristics.

2. Out of the 4 levels of parallelism discussed in lecture which one is the programmer responsible for and **why?**

Thread Level parallelism is the programmer responsible for. Only thread level parallelism is visible to programmers, for example, we can use the pthread library in C++ to explicitly make our program parallel. Others levels are either performed by compiler or done by hardware, which are out of programmers' control.

3. Which of the "2.5 architecture tricks" can adding a Cache to a processor be reduced down to? **Please explain.**

It can be reduced down to speculation with prediction.

First, cache implements the idea of speculation because loads data from memory into cache before the read request. Second, to say it's speculation with prediction is because it uses temoral and spacial locality to predict which data will be read in the near future.

4. Consider a 64-bit architecture computer executing the following code segment. How can this code be sped up using one of the "bit-level" parallelism techniques you learnt in class? **Please justify your answer.**

```
uint16_t a[100] = {0} // 16 bit integers initialized to 0
for (i = 0; i < 100; i++) {
     if (i > 50) {
          a[i] = 1;
     } else {
          a[i] += b[i]; // b is the same type and share as 'a'
and was initialized elsewhere in the program
     }
}
```

Hint: Note that up to 4 elements can be packed into a register

For the latter half of the loop where it computes `a[i] += b[i]`, we can use bit-level SIMD parallelism for speedup. Since the ALU is 64 bits, we can pack `A[i], A[i+1], A[i+2], A[i+3]` into one register and feed it to the ALU, doing the same thing for `B`. The result of one ALU operation is `A[i]+b[i], A[i+1]+B[i+1], A[i+2]+B[i+2], A[i+3]+B[i+3]`. Therefore we can get four additions done in one ALU operation, using parallel add.

Question 1

You have to compare two computers each of which implements a different ISA. For a certain benchmark, computer One, which implements ISA-A, operates at 600 MHz with an IPC of 4. Computer two, which implements ISA-B, operates at 800 MHz with an IPC of 3.

(a) What is the performance in MIPS (Millions of Instructions Per Second) of the two machines for this benchmark?

(b) Which machine has better performance? Why?

(a)
MIPS-A = 600M cycles/s * 4 instructions/cycle = 2400 MIPS
MIPS-B = 800M cycles/s * 3 instructions/cycle = 2400 MIPS

(b)
We would still like to compare performances based on CPU time.

CPU time = IC * CPI * CT

The two machines have the same CPI * CT, but we do not know the instruction count for the two machines, which depends on their ISA implementation. Therefore, the machine that has the less instruction count to execute that benchmark has the higher performance.

Question 2

Suppose we have two unpipelined machines A and B, with operating frequencies 2GHz and 2.4GHz, respectively. Both machines contain only four types of instructions: ADD, MUL, BR, and MEM. Given the delay (i.e. cycles) for each operation on the machines, and the breakdown of benchmark programs X, Y and Z by instruction type, which machine (A or B) should you buy if you care equally about the performance of all 3 programs on it? **Please show your work including the speedup of the faster machine.**

	ADD	MUL	BR	MEM
Machine A	1	3	1	4
Machine B	1	4	2	2
Program X	30%	20%	10%	40%
Program Y	20%	40%	20%	20%
Program Z	25%	25%	20%	30%

CPU time = I.C × $\overline{CPI}$ × C.T

Machine A

Program X $t_X = [0.3\times1 + 0.2\times3 + 0.1\times1 + 0.4\times4] \times \frac{1}{2GHz} = 1.3ns$

Program Y $t_Y = [0.2\times1 + 0.4\times3 + 0.2\times1 + 0.2\times4] \times \frac{1}{2GHz} = 1.2ns$

Program Z $t_Z = [0.25\times1 + 0.25\times3 + 0.2\times1 + 0.3\times4] \times \frac{1}{2GHz} = 1.2ns$

$t_A = t_X + t_Y + t_Z = 3.7ns$

Machine B

Program X $t_X = [0.3\times1 + 0.2\times4 + 0.1\times2 + 0.4\times2] \times \frac{1}{2.4G} = 0.875ns$

Program Y $t_Y = [0.2\times1 + 0.4\times4 + 0.2\times2 + 0.2\times2] \times \frac{1}{2.4G} = 1.083ns$

Program Z $t_Z = [0.25\times1 + 0.25\times4 + 0.2\times2 + 0.3\times2] \times \frac{1}{2.4G} = 0.9375ns$

$t_B = t_X + t_Y + t_Z = 2.89ns \Rightarrow$ BUY MACHINE B!

Speedup = $\frac{t_A}{t_B} = \frac{3.7}{2.89} = 1.28\times$

Question 3

We have two benchmarks, X and Y. Each benchmark is divided into multiple subtasks each with the same number of operations. The number of operations in each subtask for different benc hmarks is not necessarily the same. For benchmark X, machine A can process 2 subtasks per second, and machine B can process 3 subtasks per second. For benchmark Y, machine A can process 5 subtasks per second, and machine B can process 3 subtasks per second. If the weights of benchmark X and Y are equal, which machine would you choose? Please explain why.

We use Harmonic Mean to calculate the average rate:

Average rate A = $\frac{2}{\frac{1}{2}+\frac{1}{3}} = \frac{12}{5}$

Average rate B = $\frac{2}{\frac{1}{5}+\frac{1}{3}} = \frac{15}{4}$

Rate B > Rate A, so machine B is a better choice.

Question 4

Suppose we have a multiplication-intensive application X, and 20% of the executing time is spent by the multiplier. A smart architect optimizes the multiplier and improves the multiplication speed by 80% (1.8x).

1. What is the speedup of application X when this optimized multiplier is added?
2. If you are somehow able to further improve the multiplier's performance, what is the maximum possible speedup of X?

Speedup = (old runtime) / (new runtime) = **1 / (0.8 + 0.2/1.8) = 1.097x**

Consider the extreme case that multiplication has been optimized to a point where it takes no time to complete. The new runtime is **0.2 * 0 + 0.8 = 0.8**

The maximum possible speedup of X is **1 / 0.8 = 1.25x**